Tejas Rajendra Dapse

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.24
K.T.H.M. College, Nashik	Class XII (Maharashtra State Board of Secondary and Higher Secondary Education)	2017	89.54%
Maratha Highschool, Nashik	Class X (Maharashtra State Board of Secondary and Higher Secondary Education)	2015	91.60%

MASTER'S THESIS

Dynamic Routing Strategies for All-to-All Communication in MoE Systems.

Supervisor: Prof. Abhijit Das

- Developing dynamic routing strategies for MoE-based **distributed LLM inference/training** to reduce **All-to-All communication overhead** across interconnected GPU/NPU accelerators.
- Evaluating **2D Mesh and Torus NoC topologies** and routing algorithms to analyze **traffic distribution, network contention, link utilization, bandwidth, and communication latency**.
- Using **AstraSim and Garnet** for simulation-based **performance modeling** of MoE communication and working on **Network-on-Chip (NoC) routing algorithms** to improve communication performance.
- Tools: **AstraSim Simulator, Garnet Simulator, Python, C++.**

PROJECTS

Multi-agent AI system for manufacturing visual inspection | GitHub

- Implemented **PatchCore anomaly detection** from scratch in PyTorch; Trained on **defect-free images only** and achieved a **0.9868 mean AUROC** across **15 MVTec AD** categories.
- Cut CPU inference from 42.9 ms to 13.7 ms per image **(3.1x faster)** via an **ONNX Runtime** export, a static INT8 **post-training quantization**, and **FX graph pruning**, cutting model size from **99 MB to 25 MB (3.9x)** for **edge deployment.**
- Architected a 4-agent **LangGraph pipeline** (detect, retrieve, analyze, report) with **RAG** over a **Qdrant** vector store.
- Added a durable **human-in-the-loop** approval gate and a **SHA-256 hash-chain** audit trail for regulatory traceability.

Secure Web Payment Application | GitHub

- Designed a secure peer-to-peer payment application using **STRIDE threat modeling** and defence-in-depth architecture.
- Implemented **authentication, session management, CSRF protection, input validation,** and **audit logging.**
- Prevented double-spending using **ACID transactions, concurrency control,** and ordered row-level locking.
- **Defended** against **30 distinct attacks** with **no valid attack** landing in an 8-day exercise against 18 teams.
- Secured payment systems and transaction integrity across a hardened, network-isolated Docker deployment.

Neural Hangman Solver | Brand & Buzzword Hackathon, MeltWater | Github

- Architected a custom 6M-parameter neural network in PyTorch, fusing multi-scale **CNN, BiLSTM & Transformer** encoders.
- Devised a two-phase pipeline of 1.7M simulated states plus on-policy self-play to supervise 3 multi-task output heads.
- Achieved a 68.38% win rate on Meltwater's blind public and private test sets with zero train-test overlap and no dictionary lookup.

Systolic-Array Matrix-Multiplication Accelerator on FPGA | Group project with PhD scholar, EE Dept., IIT H | GitHub

- Designed the **RTL compute core** of an N×N **output-stationary systolic array** in **SystemVerilog**: signed int8 **MAC processing elements** with 32-bit accumulators, **input skew buffers**, and a **parameterized PE grid** run by a controller FSM.
- Derived and verified the **latency** (K + 2N cycles) and **PE utilization** (4×4, 8×8, 5.1 GOPS at 50 MHz in simulation) with **self-checking cocotb testbenches** against a NumPy golden model, using random and edge-case inputs.
- Worked in a team **co-simulation flow** (**Verilator + C++/SystemC models)**, 2,000 tests. *In progress:* **FPGA bring-up** on the Intel DE10-Lite and a **UVM** testbench.
- Tools: RTL design, cocotb, Python, Intel Quartus, DE10-Lite (MAX 10)

GeoResNet: Shape-Based Image Classification | DL Hackathon Project | GitHub

- Engineered a 5-channel geometric encoder in OpenCV, converting raw images into edge, corner and shape maps.
- Trained a **ResNet-34 CNN** in PyTorch with CBAM attention, improving focus on key shape regions.
- Applied 5-fold stratified cross-validation with AdamW and cosine scheduling, improving model generalization.
- Secured **Top 9 of 74** teams in IIT Hyderabad's Deep Learning Hackathon 2026.

RELEVANT COURSES

Advanced Data Structures & Algorithms, Foundations of Machine Learning, Deep Learning, Fraud Analytics, Visual Computing

Systems for Machine Learning, Digital Logic, Operating Systems, Computer Architecture, Computer Network, Network Security, Database Management Systems

Supply Chain Management, Operations Research

SKILLS

Languages and Database: C, C++, Java, Python, MySQL, Oracle

AI/ML: Pandas, NumPy, Scikit-learn, Statistics, PyTorch, Transformers, Mixture-of-Experts (MoE).

AI Systems & Architecture: TPU, NPU, Collective Communication (NCCL), All-to-All, NoC Routing

Simulation: ScaleSim, AstraSim, Garnet

Software Engineering: OOP, Debugging, Documentation

Developer Tools: Git, Linux, VS Code, Anti-Gravity, Claude Code

LEADERSHIP & EXTRACURRICULAR

Event Management Coordinator at MindSpark Technical Fest

- Managed technical competitions involving 300+ student participants, managing event execution, scheduling, volunteers, and participant coordination.

Regatta College Boat Club

- Participated in a competitive team-based boating event, teamwork and coordination.

SCHOLASTIC ACHIEVEMENTS & CERTIFICATES

- **Ranked 3rd** out of 1,634 students in the (12th class) Science at K.T.H.M. College, Nashik
- Honors in Mechatronics | CoEP | 2021